About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018
First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE CLASS 12 MATHEMATICS 2015

A. Formulae

Vectors and three-dimensional geometry: For vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$,

1.1.5

$$\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3, \quad \mathbf{a} \perp \mathbf{b} \iff \mathbf{a} \cdot \mathbf{b} = 0. \tag{1.1.5.1}$$

1.1.1

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}. \tag{1.1.1.1}$$

$$\text{unit vector along } \mathbf{v} = \frac{\mathbf{v}}{\|\mathbf{v}\|}, \qquad \|c\mathbf{v}\| = |c| \, \|\mathbf{v}\| \, . \tag{1.1.1.2}$$

The vector product is normal to both vectors in the product. Distances and line/plane relations are The cross product is perpendicular to both input vectors: **1.1.5**

$$\mathbf{a} \times \mathbf{b} \perp \mathbf{a}, \qquad \mathbf{a} \times \mathbf{b} \perp \mathbf{b}. \tag{1.1.5.2}$$

1.1.6

$$d(\text{parallel lines}) = \frac{\|(\mathbf{b} - \mathbf{a}) \times \mathbf{v}\|}{\|\mathbf{v}\|}. \tag{1.1.6.1}$$

1.1.2

$$d(\text{skew lines}) = \frac{|(\mathbf{b} - \mathbf{a}) \cdot (\mathbf{v}_1 \times \mathbf{v}_2)|}{\|\mathbf{v}_1 \times \mathbf{v}_2\|}. \tag{1.1.2.2}$$

1.1.10

$$\text{line parallel to plane} \iff \mathbf{d} \cdot \mathbf{n} = 0. \tag{1.1.10.1}$$

$$\text{plane through } \mathbf{r}_0 \text{ with normal } \mathbf{n}: \quad \mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0. \tag{1.1.10.2}$$

1.1.15

$$\text{line with point } \mathbf{A} \text{ and direction } \mathbf{d}: \quad \mathbf{r} = \mathbf{A} + t\mathbf{d}. \tag{1.1.15.1}$$

Matrices and determinants:

1.1.28

$$\det(AB) = \det(A)\det(B), \qquad \det(A^n) = (\det A)^n. \tag{1.1.28.1}$$

1.1.30

$$A^{-1} = \frac{\operatorname{adj}(A)}{\det A}, \qquad \det A \neq 0. \tag{1.1.30.1}$$

$$A\mathbf{x} = \mathbf{b} \implies \mathbf{x} = A^{-1}\mathbf{b} \quad (A \text{ invertible}). \tag{1.1.30.2}$$

1.1.37

$$[A \mid \mathbf{b}] \xrightarrow{\text{elementary row operations}} [I \mid \mathbf{x}] \quad \text{(Gaussian elimination).} \tag{1.1.37.1}$$

The identity used to factor the determinant in Q36 is **1.1.36**

$$a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}\big((a-b)^2 + (b-c)^2 + (c-a)^2\big). \tag{1.1.36.1}$$

Differentiation and differential equations: For $y' + P(x)y = Q(x)$, the integrating factor is **1.1.24**

$$\mu(x) = e^{\int P(x)\,dx}. \tag{1.1.24.1}$$

The logarithmic derivative is **1.1.40**

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.40.1}$$

The same identity gives **1.1.42**

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.42.1}$$

1.1.25

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}, \qquad \frac{d^2y}{dx^2} = \frac{d}{dt}(dy/dx)\Big/\frac{dx}{dt}. \tag{1.1.25.2}$$

1.1.20

$$y' + P(x)y = Q(x), \quad \mu(x) = e^{\int P(x)\,dx}, \quad \frac{d}{dx}(\mu y) = \mu Q. \tag{1.1.20.1}$$

1.1.22

$$y = vx \quad \Longrightarrow \quad y' = v + xv' \quad \text{(homogeneous first-order ODE).} \tag{1.1.22.1}$$

1.1.21

$$(1 + (y')^2)^3 = r^2(y'')^2 \quad \text{for a circle family of fixed radius } r. \tag{1.1.21.1}$$

1.1.25

$$\frac{d}{dx}\log|u(x)| = \frac{u'(x)}{u(x)}, \qquad u(x) \neq 0. \tag{1.1.25.1}$$

Integration and area: A useful definite-integral antiderivative is **1.1.39**

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a}\log\left|\frac{a+u}{a-u}\right| + C. \tag{1.1.39.1}$$

After polynomial division, partial fractions can be written as **1.1.43**

$$\frac{P(x)}{(x-a)(x^2+b^2)} = Q(x) + \frac{A}{x-a} + \frac{Bx+C}{x^2+b^2}. \tag{1.1.43.1}$$

The trigonometric integrand in Question 44 can be split using **1.1.44**

$$\frac{5\sin x + 3\cos x}{\sin x + \cos x} = 4 + \frac{\sin x - \cos x}{\sin x + \cos x}. \tag{1.1.44.1}$$

Integration by parts states **1.1.45**

$$\int u\,dv = uv - \int v\,du. \tag{1.1.45.2}$$

Area between curves is **1.1.47**

$$A = \int_a^b |y_{\text{upper}} - y_{\text{lower}}|\,dx. \tag{1.1.47.1}$$

1.1.38

$$\int \frac{du}{a^2 - u^2} = \frac{1}{2a}\log\left|\frac{a+u}{a-u}\right| + C. \tag{1.1.38.1}$$

$$\int \frac{du}{1+u^2} = \tan^{-1} u + C. \tag{1.1.38.2}$$

1.1.46

$$\int \frac{u\,du}{\sqrt{1+u^2}} = \sqrt{1+u^2} + C. \tag{1.1.46.2}$$

$$A = \int_a^b |y_{\text{upper}} - y_{\text{lower}}|\,dx. \tag{1.1.46.1}$$

1.1.41

$$\frac{P(x)}{(x-a)(x^2+b^2)} = Q(x) + \frac{A}{x-a} + \frac{Bx+C}{x^2+b^2} \quad \text{(partial fractions after division).} \tag{1.1.41.1}$$

1.1.45

$$\int u\,dv = uv - \int v\,du. \tag{1.1.45.1}$$

Functions, relations and operations: Continuity at an interior point requires matching one-sided limits: **1.1.52**

$$\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x). \tag{1.1.52.1}$$

At an interior point, continuity requires matching one-sided limits; differentiability additionally requires matching one-sided derivatives: **1.1.48**

$$\lim_{x\to a^-} f(x) = f(a) = \lim_{x\to a^+} f(x), \qquad f'_-(a) = f'_+(a) = f'(a). \tag{1.1.48.1}$$

1.1.49

$$\begin{aligned} &\text{reflexive: } (a,a) \in R, \\ &\text{symmetric: } (a,b) \in R \Rightarrow (b,a) \in R, \\ &\text{transitive: } (a,b),(b,c) \in R \Rightarrow (a,c) \in R. \end{aligned} \tag{1.1.49.1}$$

1.1.50

For componentwise addition on $\mathbb{R}^2$: $\quad e=(0,0), \quad (a,b)^{-1}=(-a,-b).$ (1.1.50.1)

1.1.51

$a*b=(a+b) \bmod n, \quad 0$ is the identity, $\quad (-a) \bmod n$ is the additive inverse. (1.1.51.1)

Probability: For a binomial random variable, **1.1.56**

$$P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}. \tag{1.1.56.1}$$

The law of total probability is **1.1.57**

$$P(E)=\sum_i P(E \mid A_i)P(A_i). \tag{1.1.57.1}$$

Bayes' theorem is **1.1.58**

$$P(A \mid E)=\frac{P(E \mid A)P(A)}{P(E)}. \tag{1.1.58.2}$$

1.1.58

$$P(A \mid E)=\frac{P(A \cap E)}{P(E)}. \tag{1.1.58.1}$$

1.1.54

$$P(A \mid E)=\frac{P(E \mid A)P(A)}{P(E)}. \tag{1.1.54.1}$$

1.1.53

$$X \sim \mathrm{Bin}(n,p): \quad P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}. \tag{1.1.53.1}$$

$$G(z)=\sum_j P(D=j)z^j. \tag{1.1.53.2}$$

$G_5(z)=(0.4z+0.6z^{-1})^5$ for the five-step walk.

1.1.55

$$E[X]=\sum_x xP(X=x), \quad \mathrm{Var}(X)=E[X^2]-E[X]^2. \tag{1.1.55.1}$$

$$P(X=k)=\frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}. \tag{1.1.55.2}$$

1.1.54

$$P(E)=\sum_i P(E \mid A_i)P(A_i). \tag{1.1.54.2}$$

Linear programming: For a linear programme, compare the objective at every feasible vertex: **1.1.60**

$$\text{LPP optimum} = \text{best objective value among feasible vertices.} \tag{1.1.60.1}$$

1.1.59

Evaluate the objective at each feasible corner; an optimum occurs at a feasible vertex. (1.1.59.1)

Coordinate geometry and optimisation: The tangent to $x^2 + y^2 = r^2$ at $(r\cos\theta, r\sin\theta)$ is **1.1.64**

$$x\cos\theta + y\sin\theta = r. \tag{1.1.64.1}$$

For the ellipse $x^2/a^2 + y^2/b^2 = 1$, a parametrisation is $(a\cos t, b\sin t)$, and triangle area is one half base times perpendicular height: **1.1.65**

$$\text{ellipse point } (a\cos t, b\sin t), \quad A_{\triangle} = \frac{1}{2}bh. \tag{1.1.65.1}$$

Engineering methods: choosing the right tool: Matrix methods are used for vector and matrix conditions and for Gaussian elimination. The five-step random walk and the small binomial problem use probability-generating polynomial coefficients where it makes the probability structure clearer. Laplace transforms are useful for suitable linear time-domain differential equations with initial conditions; they are not forced into the nonlinear or variable-coefficient differential equations in this set where they would not simplify the solution.

B. Problems

1.2.1 Find a vector of magnitude $\sqrt{171}$ perpendicular to both $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ -3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 3 \\ -1 \\ -2 \end{pmatrix}$.

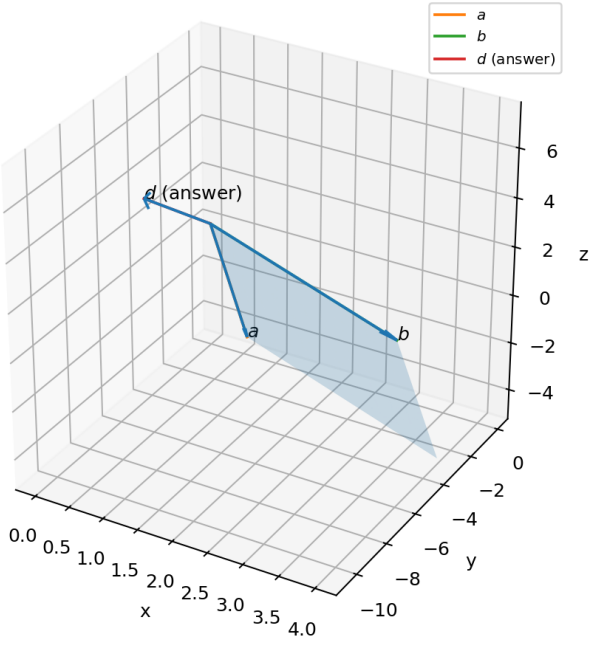
Solution: A vector perpendicular to both is parallel to their cross product:

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} 1 \\ -7 \\ 5 \end{pmatrix}, \qquad \|\mathbf{a} \times \mathbf{b}\| = \sqrt{75} = 5\sqrt{3} \qquad (1.1.1.1). \qquad (1.2.1.1)$$

Scale this direction to the required magnitude:

$$\mathbf{d} = \frac{\sqrt{171}}{5\sqrt{3}} \begin{pmatrix} 1 \\ -7 \\ 5 \end{pmatrix} = \pm \frac{\sqrt{57}}{5} \begin{pmatrix} 1 \\ -7 \\ 5 \end{pmatrix}.$$

Q1: the answer vector is normal to the plane spanned by a and b



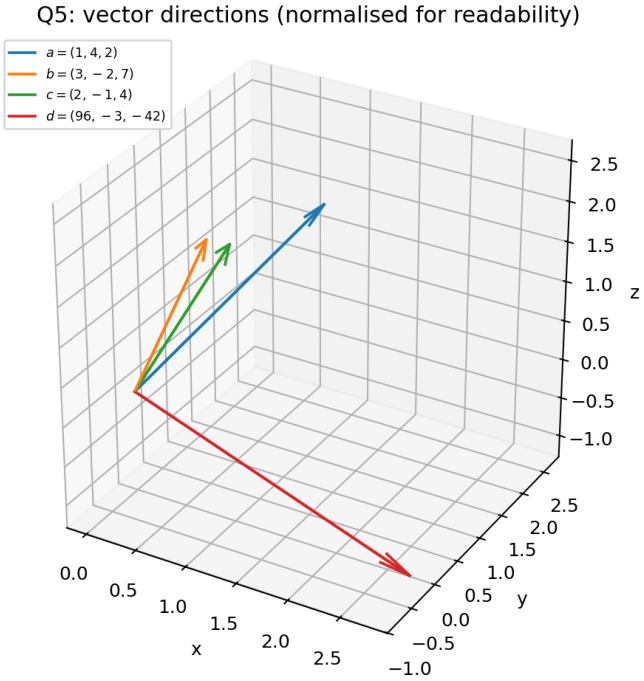
Code reference -

`codes/Q01.py`

1.2.5 Let $\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} 3 \\ -2 \\ 7 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$. Find $\mathbf{d}$ perpendicular to both $\mathbf{a}, \mathbf{b}$ with $\mathbf{c} \cdot \mathbf{d} = 27$. **Solution:** The required direction is $\mathbf{a} \times \mathbf{b} = (32, -1, -14)^T$. Set $\mathbf{d} =$

$t(32, -1, -14)^T$. Since $\mathbf{c} \cdot (\mathbf{a} \times \mathbf{b}) = 9$, the constraint gives $9t = 27$, hence

$$\mathbf{d} = \begin{pmatrix} 96 \\ -3 \\ -42 \end{pmatrix} \qquad (1.1.1.1).$$



Code reference -

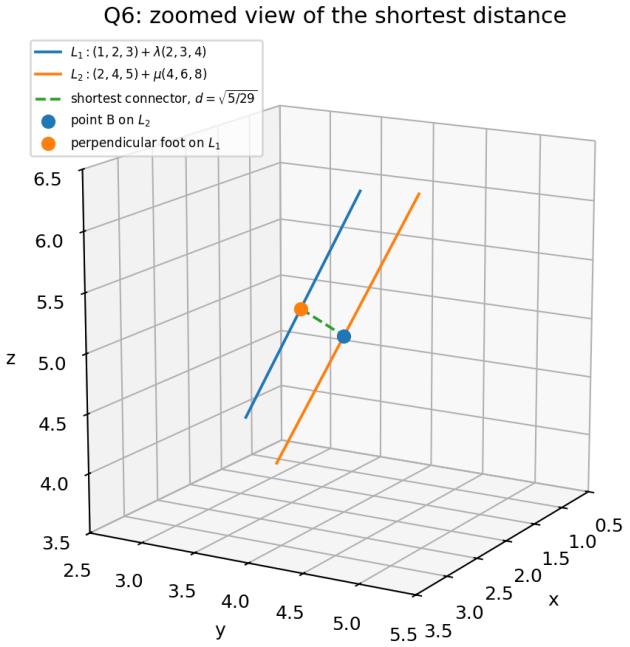
codes/Q05.py

1.2.6 Find the shortest distance between $\mathbf{r} = (1, 2, 3) + \lambda(2, 3, 4)$ and $\mathbf{r} = (2, 4, 5) + \mu(4, 6, 8)$.
Solution: The direction vectors are parallel, since $(4, 6, 8) = 2(2, 3, 4)$. For parallel lines the distance is the perpendicular distance from one point to the other line:

$$d = \frac{\|(B - A) \times \mathbf{v}\|}{\|\mathbf{v}\|}, \quad B - A = (1, 2, 2), \quad \mathbf{v} = (2, 3, 4).$$

Here $(B - A) \times \mathbf{v} = (2, 0, -1)$, so

$$d = \frac{\sqrt{5}}{\sqrt{29}} \qquad (1.1.6.1).$$



Code reference -

codes/Q06.py

1.2.10 Find the plane through the intersection of $2x + y - z = 3$ and $5x - 3y + 4z + 9 = 0$ that is parallel to $\frac{x-1}{2} = \frac{y-3}{-4} = \frac{5-z}{-5}$. **Solution:** A plane through the intersection has equation

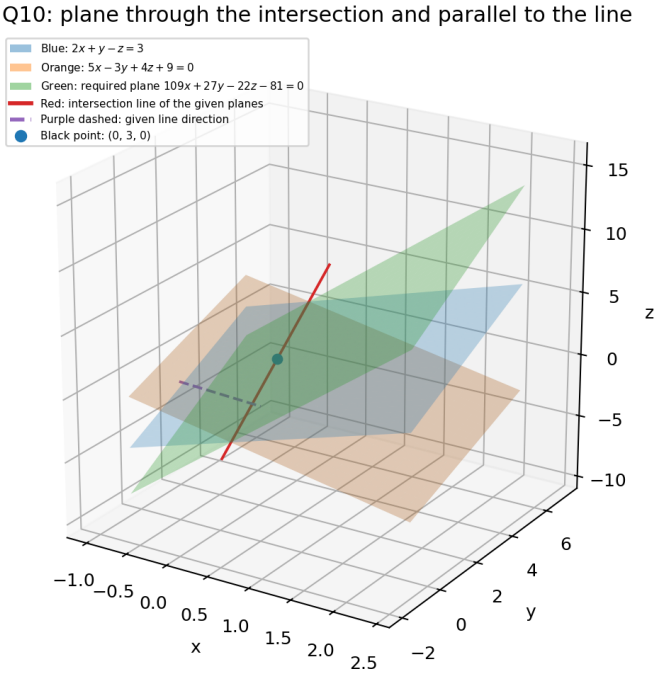
$$(2x + y - z - 3) + \lambda(5x - 3y + 4z + 9) = 0.$$

The line direction is $(2, -4, 5)$. Its dot product with the plane normal $(2+5\lambda, 1-3\lambda, -1+4\lambda)$ must vanish:

$$2(2 + 5\lambda) - 4(1 - 3\lambda) + 5(-1 + 4\lambda) = 0,$$

so $\lambda = 5/42$. Multiplying by 42 and simplifying gives

$$109x + 27y - 22z - 81 = 0 \qquad (1.1.10.1),\ (1.1.10.2).$$

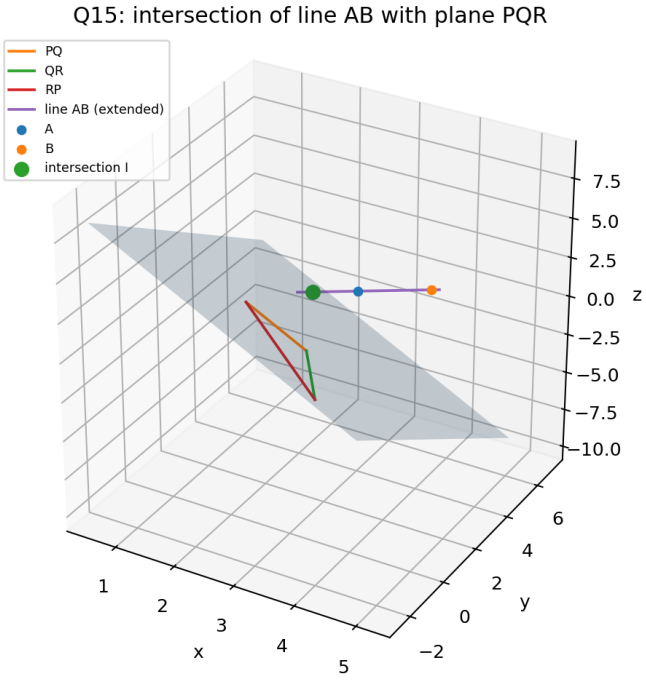


Code reference -

codes/Q10.py

1.2.15 Find where the line through $A(3,4,1)$ and $B(5,1,6)$ meets the plane through $P(2,1,2)$, $Q(3,1,0)$ and $R(4,-2,1)$. **Solution:** A normal to the plane is $(Q-P)\times(R-P) = (1,0,-2)\times(2,-3,-1) \propto (2,1,1)$. Thus the plane is $2x+y+z=7$. The line is $(3,4,1)+t(2,-3,5)$. Substitution into the plane yields $4+6t=0$, so $t=-2/3$. The intersection is

$$\left(\frac{5}{3}, 6, -\frac{7}{3}\right) \qquad (1.1.15.1),\ (1.1.10.2).$$



Code reference -

`codes/Q15.py`

1.2.20 Find the integrating factor of $x \log x \, \frac{dy}{dx} + y = 2 \log x$. **Solution:** In standard form,

$$y' + \frac{1}{x \log x} y = \frac{2}{x}.$$

Thus the integrating factor is

$$\exp\left(\int \frac{dx}{x \log x} \right) = \exp(\log |\log x|) \qquad (1.1.24.1).$$

On a domain where $\log x > 0$, this is $\log x$ (more generally the factor can be taken as $|\log x|$ on an interval not crossing $x = 1$).

1.2.21 Find the differential equation of the family $(x-h)^2 + (y-k)^2 = r^2$, where h, k are arbitrary constants and r is fixed. **Solution:** Differentiate once:

$$(x-h) + (y-k)y' = 0.$$

Differentiate again: $1 + (y')^2 + (y-k)y'' = 0$, giving $y - k = -(1 + (y')^2)/y''$. Substitute

this and $x-h=-(y-k)y'$ into the original circle equation. Then

$$\frac{(1+(y')^2)^3}{(y'')^2}=r^2, \quad (1+(y')^2)^3=r^2(y'')^2 \qquad (1.1.21.1).$$

1.2.22 Show that $(x-y)\frac{dy}{dx}=x+2y$ is homogeneous and solve it. **Solution:** The derivative is a ratio of homogeneous first-degree expressions. Put $y=vx$, so $y'=v+xv'$. Substitution gives

$$x\frac{dv}{dx}=\frac{1+v+v^2}{1-v}, \qquad \frac{1-v}{1+v+v^2}\,dv=\frac{dx}{x} \qquad (1.1.22.1).$$

Integrating,

$$-\frac{1}{2}\ln(v^2+v+1)+\sqrt{3}\tan^{-1}\left(\frac{2v+1}{\sqrt{3}}\right)=\ln|x|+C.$$

Using $v=y/x$ gives the implicit solution

$$-\tfrac{1}{2}\ln\left(\frac{y^2}{x^2}+\frac{y}{x}+1\right)+\sqrt{3}\tan^{-1}\left(\frac{2y/x+1}{\sqrt{3}}\right)=\ln|x|+C.$$

1.2.24 Find the particular solution of $y'+y\tan x=3x^2+x^3\tan x$, given $y=0$ at $x=\pi/3$ and $x\neq\pi/2$. **Solution:** The integrating factor is $e^{\int\tan x\,dx}=\sec x$. Multiplication gives

$$\frac{d}{dx}(y\sec x)=3x^2\sec x+x^3\sec x\tan x=\frac{d}{dx}(x^3\sec x) \qquad (1.1.24.1).$$

So $y\sec x=x^3\sec x+C$ and $y=x^3+C\cos x$. From $y(\pi/3)=0$, $C=-2\pi^3/27$. Therefore

$$y=x^3-\frac{2\pi^3}{27}\cos x.$$

Code reference -

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codes/Q24.py
```

1.2.25 If $y=\log\left(\frac{x}{a+bx}\right)^x$, prove that $x^3y''=(xy'-y)^2$. **Solution:** Interpret the expression as $y=x\log\left(\frac{x}{a+bx}\right)$. Then

$$y'=\log\left(\frac{x}{a+bx}\right)+\frac{a}{a+bx}, \qquad y''=\frac{a^2}{x(a+bx)^2} \qquad (1.1.40.1).$$

Thus $xy'-y=ax/(a+bx)$ and

$$x^3y''=\frac{a^2x^2}{(a+bx)^2}=(xy'-y)^2.$$

1.2.28 If $A=\begin{pmatrix}\cos\theta & \sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}$, find $\det(A^n)$ for natural n. **Solution:** Determinants are multiplicative, so $\det(A^n)=(\det A)^n$. Now $\det A=\cos^2\theta-\sin^2\theta=\cos 2\theta$. Thus

$$\det(A^n)=(\cos 2\theta)^n \qquad (1.1.28.1).$$

1.2.30 Find the inverse of $\begin{pmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 0 \end{pmatrix}$ using elementary row operations. **Solution:** Augment the matrix by I and row-reduce to $[I \mid A^{-1}]$. The determinant is -1, so the matrix is invertible. The result is

$$A^{-1} = \begin{pmatrix} 3 & -2 & 1 \\ -9 & 6 & -2 \\ 5 & -3 & 1 \end{pmatrix} \qquad (1.1.30.1).$$

The product check $AA^{-1} = I$ is carried out in code.

Code reference -

codes/Q30.py

1.2.36 If $a+b+c \neq 0$ and $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$, prove that $a = b = c$. **Solution:** Expanding and factorising the determinant gives

$$-(a+b+c)(a^2+b^2+c^2-ab-bc-ca) = 0.$$

The first factor is nonzero by assumption, so $a^2+b^2+c^2-ab-bc-ca = 0$. But

$$a^2+b^2+c^2-ab-bc-ca = \tfrac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2] \qquad (1.1.36.1).$$

Each square must be zero, hence $a = b = c$.

1.2.37 A trust receives Rs. 30,000 per month. Half is spent on care, and 2% of the amount spent is collected from beneficiaries. The unspent balance is deposited in a bank. Find the monthly interest rate needed for total monthly income of Rs. 1800. **Solution:** Let u be the monthly contribution collected from beneficiaries and v the bank interest. Half of Rs. $30,000$ is spent, so

$$u = 0.02(15000) = 300.$$

The total monthly income is Rs. 1800, so

$$u + v = 1800, \qquad u = 300.$$

Write this system as an augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 1 & 0 & 300 \end{array}\right] \xrightarrow{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 0 & -1 & -1500 \end{array}\right] \xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{cc|c} 1 & 1 & 1800 \\ 0 & 1 & 1500 \end{array}\right].$$

Eliminate v from the first row:

$$R_1 \leftarrow R_1 - R_2: \qquad \left[\begin{array}{cc|c} 1 & 0 & 300 \\ 0 & 1 & 1500 \end{array}\right] \qquad (1.1.37.1).$$

Thus $u = 300$ and $v = 1500$. The deposit is Rs. 15000, therefore

$$r = \frac{1500}{15000} \times 100\% = 10\% \text{ per month.}$$

Code reference -

`codes/Q37.py`

1.2.38 Evaluate $I = \displaystyle\int_0^{\pi/2} \frac{\cos^2 x}{1+3\sin^2 x}\,dx$. **Solution:** Let $t = \tan x$. Then $dx = dt/(1+t^2)$ and the integral becomes

$$I = \int_0^{\infty} \frac{dt}{(1+t^2)(1+4t^2)} = \int_0^{\infty} \left[-\frac{1}{3(1+t^2)} + \frac{4}{3(1+4t^2)}\right] dt.$$

Evaluating the arctangent limits gives $I = -\pi/6 + \pi/3 = \pi/6$.

1.2.39 Evaluate $\displaystyle\int_0^{\pi/4} \frac{\sin x + \cos x}{3+\sin 2x}\,dx$. **Solution:** Set $u = \sin x - \cos x$, so $du = (\sin x + \cos x)dx$ and $\sin 2x = 1 - u^2$. The limits become $u = -1$ and $u = 0$. Hence

$$I = \int_{-1}^{0} \frac{du}{4-u^2} = \left[\frac{1}{4}\ln\left|\frac{2+u}{2-u}\right|\right]_{-1}^{0} = \frac{1}{4}\ln 3 \qquad (1.1.39.1).$$

1.2.40 Evaluate $\displaystyle\int \frac{\sin x - x\cos x}{x(x+\sin x)}\,dx$. **Solution:** Since

$$\frac{d}{dx}\ln\left|\frac{x}{x+\sin x}\right| = \frac{\sin x - x\cos x}{x(x+\sin x)} \qquad (1.1.42.1),$$

the integral is $\ln\left|\frac{x}{x+\sin x}\right| + C$.

1.2.41 Evaluate $\displaystyle\int \frac{x^3}{(x-1)(x^2+1)}\,dx$. **Solution:** Polynomial division and partial fractions give

$$\frac{x^3}{(x-1)(x^2+1)} = 1 + \frac{1}{2(x-1)} + \frac{x-1}{2(x^2+1)} \qquad (1.1.43.1).$$

Therefore

$$x + \frac{1}{2}\ln|x-1| + \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1}x + C.$$

1.2.42 Find $\displaystyle\int \frac{x\,dx}{1+x\tan x}$. **Solution:** Rewrite the integrand as $x\cos x/(\cos x + x\sin x)$. Since $d(\cos x + x\sin x)/dx = x\cos x$,

$$\int \frac{x\,dx}{1+x\tan x} = \ln|\cos x + x\sin x| + C.$$

1.2.43 Evaluate $\displaystyle\int \frac{x^4}{(x-1)(x^2+1)}\,dx$. **Solution:** Division gives $x^4/[(x-1)(x^2+1)] = x + 1 + 1/[(x-1)(x^2+1)]$. Decompose the last term as

$$\frac{1}{(x-1)(x^2+1)} = \frac{1}{2(x-1)} - \frac{x+1}{2(x^2+1)} \qquad (1.1.43.1).$$

Thus

$$\frac{x^2}{2} + x + \frac{1}{2}\ln|x-1| - \frac{1}{4}\ln(x^2+1) - \frac{1}{2}\tan^{-1}x + C.$$

1.2.44 Evaluate $\displaystyle\int_0^{\pi/2} \frac{5\sin x + 3\cos x}{\sin x + \cos x}\,dx$. **Solution:** Write the numerator as $4(\sin x + \cos x) + (\sin x - \cos x)$. Then

$$I = 4\int_0^{\pi/2} dx - [\ln(\sin x + \cos x)]_0^{\pi/2} = 2\pi \qquad (1.1.44.1).$$

1.2.45 Find $\displaystyle\int \left[\log(\log x) + \frac{1}{(\log x)^2}\right] dx$. **Solution:** Differentiate $x\log(\log x) - x/\log x$ to check that it equals the integrand. Hence

$$x\log(\log x) - \frac{x}{\log x} + C, \qquad x > 1 \text{ for a real-valued } \log(\log x).$$

1.2.46 If the area bounded by $y^2 = 16ax$ and $y = 4mx$ is $a^2/2$, find m. **Solution:** Use the upper branch $y = 4\sqrt{ax}$. At an intersection with $y = 4mx$,

$$4\sqrt{ax} = 4mx \quad\Longrightarrow\quad x = 0 \quad \text{or} \quad x = \frac{a}{m^2}.$$

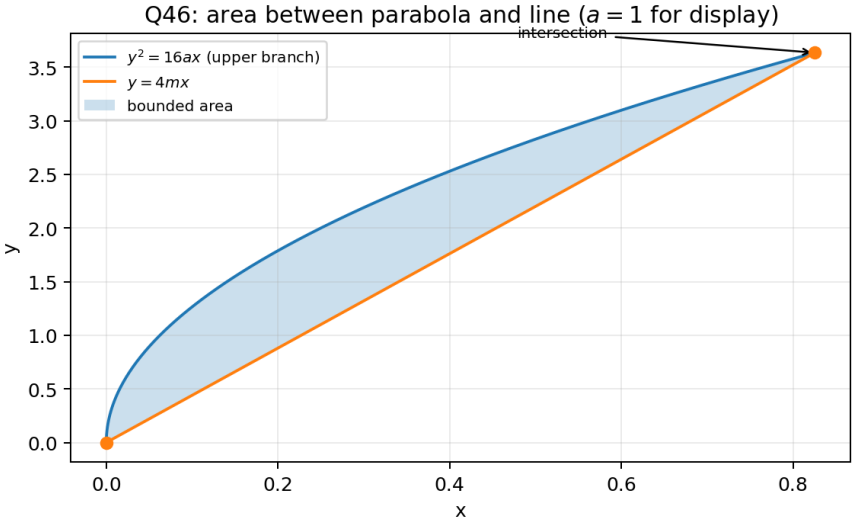
For positive slope, the area is the integral of the parabola minus the line:

$$\begin{aligned} A &= \int_0^{a/m^2} \left(4\sqrt{ax} - 4mx\right)\,dx \\ &= \left[\frac{8\sqrt{a}}{3}x^{3/2} - 2mx^2\right]_0^{a/m^2} \\ &= \frac{8a^2}{3m^3} - \frac{2a^2}{m^3} = \frac{2a^2}{3m^3} \qquad (1.1.47.1). \end{aligned}$$

Set the result equal to the given area:

$$\frac{2a^2}{3m^3} = \frac{a^2}{2} \quad\Longrightarrow\quad 4 = 3m^3 \quad\Longrightarrow\quad m = \sqrt[3]{\frac{4}{3}}.$$

The graph uses $a = 1$ for visibility; the value of m is independent of that scale.



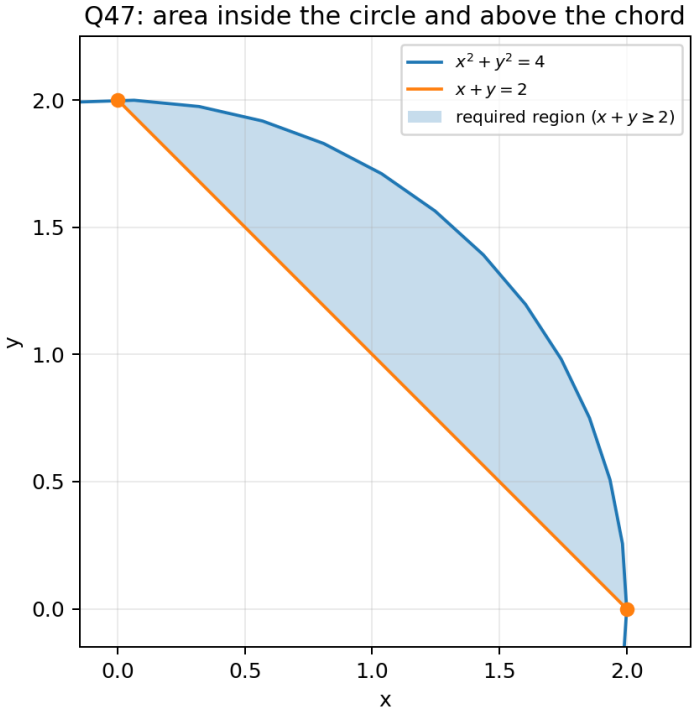
Code reference -

`codes/Q46.py`

1.2.47 Find the area of $\{(x, y) : x^2 + y^2 \leq 4,\ x + y \geq 2\}$ by integration. **Solution:** The chord $x + y = 2$ meets the circle at $(0, 2)$ and $(2, 0)$. For $0 \leq x \leq 2$, the upper circle boundary is $y = \sqrt{4 - x^2}$ and the line is $y = 2 - x$. Therefore

$$\begin{aligned} A &= \int_0^2 \left(\sqrt{4 - x^2} - (2 - x)\right) dx \\ &= \underbrace{\int_0^2 \sqrt{4 - x^2}\, dx}_{\text{quarter-circle area } = \pi} - \underbrace{\int_0^2 (2 - x)\, dx}_{\text{triangle area } = 2} \\ &= \pi - 2 \quad \text{square units} \qquad (1.1.47.1). \end{aligned}$$

The graph shades the circular segment above the chord.



Code reference -

`codes/Q47.py`

1.2.48 Discuss continuity and differentiability of $f(x) = |x| + |x-1|$ on $(-1, 2)$. **Solution:** First split the absolute values at $x = 0$ and $x = 1$:

$$f(x) = \begin{cases} 1 - 2x, & -1 < x < 0, \\ 1, & 0 \leq x < 1, \\ 2x - 1, & 1 \leq x < 2. \end{cases}$$

Continuity. At $x = 0$, the left limit, right limit and function value are all 1. The same is true at $x = 1$. Each branch is continuous on its interval; hence f is continuous throughout $(-1, 2)$.

Differentiability at $x = 0$. The left-hand slope is -2 and the right-hand slope is 0:

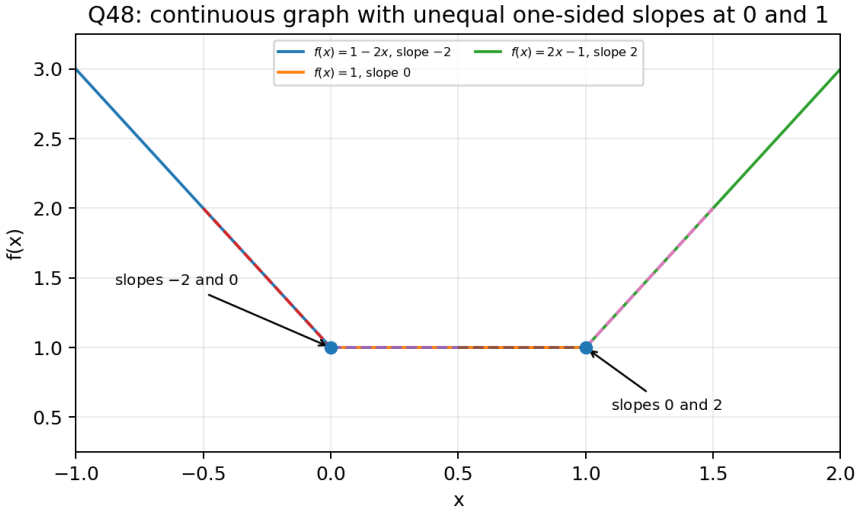
$$f'_-(0) = -2 \neq 0 = f'_+(0).$$

Thus f is not differentiable at 0.

Differentiability at $x = 1$. The left-hand slope is 0 and the right-hand slope is 2:

$$f'_-(1) = 0 \neq 2 = f'_+(1).$$

Thus f is not differentiable at 1. At other points the function is linear and differentiable. The plot shows these one-sided slopes: differentiability requires equal one-sided slopes, and the two corners fail that test.



Code reference -

`codes/Q48.py`

1.2.49 On $\mathbb{R}$, define $(a, b) \in R$ when $a - b + \sqrt{3}$ is irrational. Determine whether R is reflexive, symmetric and transitive. **Solution:** We test each property separately.

1. Reflexive. Reflexivity requires $(a, a) \in R$ for every real a. Substituting $b = a$ gives

$$a - a + \sqrt{3} = \sqrt{3},$$

which is irrational. Therefore R is reflexive.

2. Symmetric. Symmetry requires $(a, b) \in R$ to imply $(b, a) \in R$. Choose $a = \sqrt{3}$ and $b = 0$. Then

$$a - b + \sqrt{3} = 2\sqrt{3}$$

is irrational, so $(a, b) \in R$. Reversing the pair gives

$$b - a + \sqrt{3} = 0,$$

which is rational, so $(b, a) \notin R$. Hence R is not symmetric.

3. Transitive. Transitivity requires $(a, b), (b, c) \in R$ to imply $(a, c) \in R$. Choose $a = \sqrt{3}$, $b = 0$, and $c = 2\sqrt{3}$. Then

$$a - b + \sqrt{3} = 2\sqrt{3}, \qquad b - c + \sqrt{3} = -\sqrt{3},$$

both irrational, so $(a,b),(b,c)\in R$. However,

$$a-c+\sqrt{3}=\sqrt{3}-2\sqrt{3}+\sqrt{3}=0,$$

which is rational. Thus $(a,c)\notin R$, and R is not transitive.
Conclusion. R is reflexive, but neither symmetric nor transitive.

1.2.50 On $\mathbb{R}\times\mathbb{R}$, define $(a,b)*(c,d)=(a+c,b+d)$. Prove commutativity and associativity, and find the identity and the inverse of $(3,-5)$. **Solution:** Addition of real components is commutative and associative, so $*$ has both properties. Its identity is $(0,0)$, since $(a,b)*(0,0)=(a,b)$. The inverse of $(3,-5)$ is $(-3,5)$.

1.2.51 On $\{0,1,2,3,4,5,6\}$ define $a*b=a+b$ if $a+b<7$, and $a*b=a+b-7$ otherwise. Give the operation table and show that 0 is the identity and $7-a$ is the inverse for nonzero a. **Solution:** This is addition modulo 7. Its table is

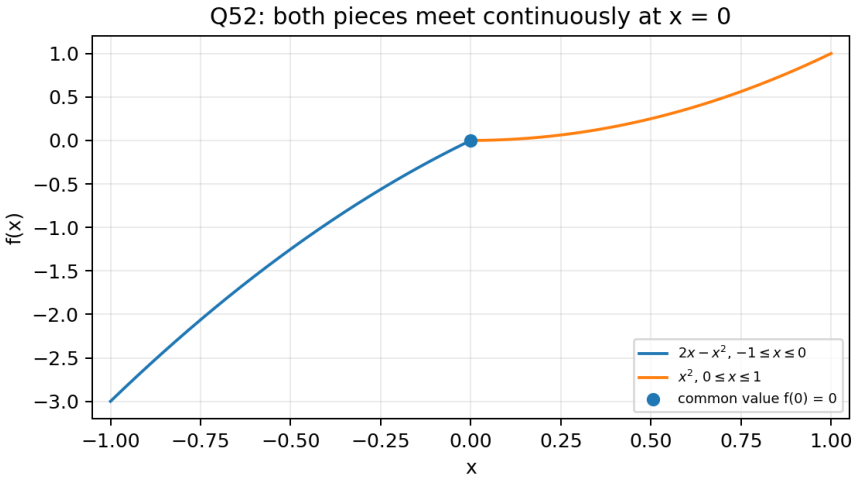
$*$	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

Row and column 0 reproduce every element, hence 0 is the identity. For each $a=1,\dots,6$, $a+(7-a)=7\equiv 0 \pmod 7$, so $7-a$ is its inverse.

1.2.52 Find any point of discontinuity of $f(x)=x-|x-x^2|$, $x\in[-1,1]$. **Solution:** On $[-1,1]$, $x-x^2=x(1-x)$, whose sign changes only at 0. Thus

$$f(x)=\begin{cases}2x-x^2, & -1\le x\le 0,\\ x^2, & 0\le x\le 1.\end{cases}$$

Both pieces take value 0 at $x=0$ and meet continuously. The endpoints are also included continuously. Therefore there are no points of discontinuity.



Code reference -

codes/Q52.py

1.2.53 A man steps forward with probability 0.4 and backward with probability 0.6. Find the probability that after 5 steps he is one step away from his starting point. **Solution:** Let k be the number of forward steps. The net displacement after five steps is

$$D = k - (5 - k) = 2k - 5.$$

One step away means $D = +1$ or $D = -1$. These occur when $k = 3$ or $k = 2$, respectively.

Step 1: Net displacement $+1$. Three steps are forward and two backward, so the binomial formula gives

$$\begin{aligned} P(D = 1) &= \binom{5}{3}(0.4)^3(0.6)^2 \\ &= 10(0.064)(0.36) = 0.2304 \qquad (1.1.53.1). \end{aligned}$$

Step 2: Net displacement -1. Two steps are forward and three backward:

$$\begin{aligned} P(D = -1) &= \binom{5}{2}(0.4)^2(0.6)^3 \\ &= 10(0.16)(0.216) = 0.3456. \end{aligned}$$

Step 3: Add the disjoint events.

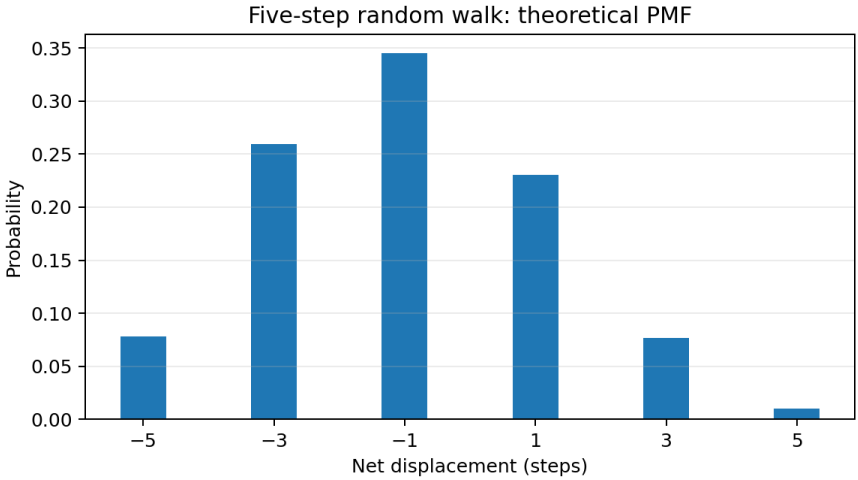
$$P(|D| = 1) = P(D = 1) + P(D = -1) = 0.2304 + 0.3456 = 0.576.$$

Generating function / Z-transform. For one step, the Laurent probability generating

function is $G(z) = 0.4z + 0.6z^{-1}$. For five independent steps,

$$G_5(z) = (0.4z + 0.6z^{-1})^5 = \sum_{k=0}^{5} \binom{5}{k} (0.4)^k (0.6)^{5-k} z^{2k-5} \qquad (1.1.53.2).$$

The coefficient of z^1 is 0.2304 and the coefficient of z^{-1} is 0.3456. Their sum is the required probability. The plot shows the theoretical PMF for every possible displacement.



Code reference -

`codes/Q53.py`

1.2.54 A girl throws a die. If it shows 1 or 2, she tosses a coin three times; otherwise she tosses it once. Given that exactly one tail was obtained, find the probability that the die showed 3, 4, 5 or 6. **Solution:** Let A mean that the die shows 1 or 2, B mean that it shows 3, 4, 5 or 6, and E mean that exactly one tail is observed.

Step 1: Find the prior probabilities.

$$P(A) = \frac{2}{6} = \frac{1}{3}, \qquad P(B) = \frac{4}{6} = \frac{2}{3}.$$

Step 2: Find the likelihood of E. If A occurs, three fair coin tosses are made. Exactly one tail can occur in $\binom{3}{1}$ of the 8 equally likely outcomes:

$$P(E \mid A) = \binom{3}{1} \left(\frac{1}{2}\right)^3 = \frac{3}{8}.$$

If B occurs, only one coin toss is made, so

$$P(E \mid B) = \frac{1}{2}.$$

Step 3: Calculate $P(E)$ by total probability.

$$\begin{aligned} P(E) &= P(E \mid A)P(A) + P(E \mid B)P(B) \\ &= \frac{3}{8}\cdot\frac{1}{3} + \frac{1}{2}\cdot\frac{2}{3} \\ &= \frac{1}{8} + \frac{1}{3} = \frac{11}{24} \qquad (1.1.57.1). \end{aligned}$$

Step 4: Apply Bayes' theorem.

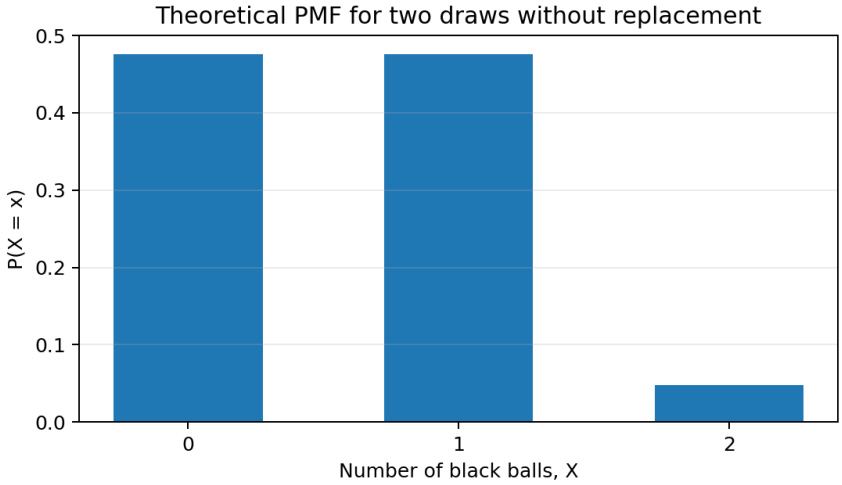
$$P(B \mid E) = \frac{P(E \mid B)P(B)}{P(E)} = \frac{(1/2)(2/3)}{11/24} = \frac{8}{11} \qquad (1.1.58.2).$$

1.2.55 An urn contains 5 red and 2 black balls. Two balls are drawn without replacement. Let X be the number of black balls. State its possible values and find its mean and variance. **Solution:** The possible values are $0, 1, 2$. The probabilities from the hypergeometric model are

$$P(X=0) = \frac{10}{21}, \quad P(X=1) = \frac{10}{21}, \quad P(X=2) = \frac{1}{21} \qquad (1.1.55.2).$$

So $E[X] = 0(10/21) + 1(10/21) + 2(1/21) = 4/7$. Also $E[X^2] = 10/21 + 4/21 = 2/3$, hence

$$\mathrm{Var}(X) = E[X^2] - E[X]^2 = \frac{2}{3} - \frac{16}{49} = \frac{50}{147} \qquad (1.1.55.1).$$



Code reference -

codes/Q55.py

1.2.56 In 3 trials of a binomial distribution, the probability of exactly 2 successes is 9 times the probability of 3 successes. Find the success probability. **Solution:** Let the success

probability be p and $q = 1 - p$. The probability-generating polynomial is

$$B(z) = (q + pz)^3 = q^3 + 3pq^2 z + 3p^2 q z^2 + p^3 z^3.$$

The coefficient of z^2 is

$$P(X = 2) = \binom{3}{2} p^2 q = 3p^2 q \qquad (1.1.56.1),$$

and the coefficient of z^3 is

$$P(X = 3) = p^3.$$

Use $P(X = 2) = 9P(X = 3)$:

$$3p^2 q = 9p^3,$$
$$3p^2(1 - p) = 9p^3.$$

For the non-degenerate binomial case $p > 0$, divide by $3p^2$: $1 - p = 3p$. Thus $4p = 1$ and

$$p = \frac{1}{4}.$$

The polynomial's coefficients directly identify the two probabilities; no separate Z-transform is needed beyond this generating-polynomial viewpoint.

Code reference -

codes/Q56.py

1.2.57 An urn contains 3 red and 5 black balls. A ball is drawn, its colour recorded and returned, and 2 more balls of that colour are added. Then two balls are drawn without replacement. Find the probability both are red. **Solution:** If the first ball is red, with probability $3/8$, the urn becomes 5 red and 5 black; the chance of two reds is $(5/10)(4/9) = 2/9$. If the first ball is black, with probability $5/8$, the urn becomes 3 red and 7 black; the chance of two reds is $(3/10)(2/9) = 1/15$. Thus

$$P(\text{two red}) = \frac{3}{8}\frac{2}{9} + \frac{5}{8}\frac{1}{15} = \frac{1}{8} \qquad (1.1.57.1).$$

Code reference -

codes/Q57.py

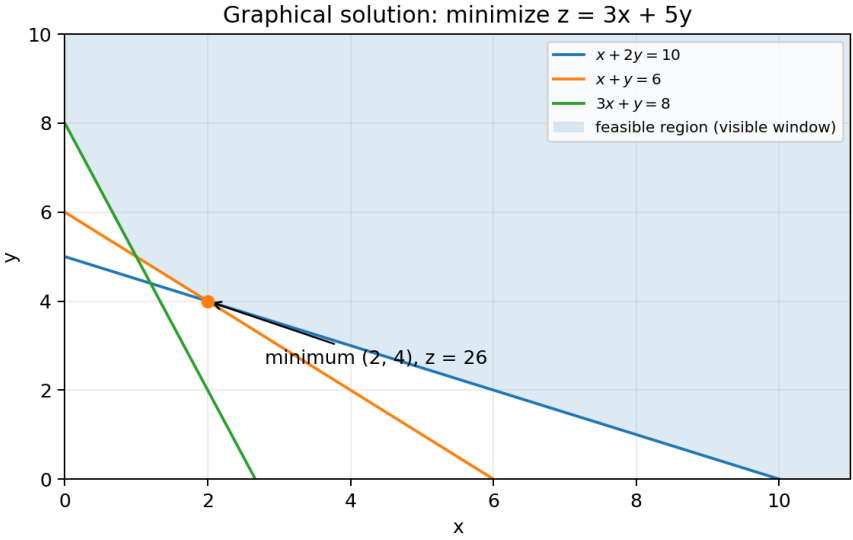
1.2.58 A man speaks truth 3 out of 5 times. He throws a die and reports that it is 4. Find the probability that it actually is 4. **Solution:** Let A be the event that the die shows 4, and R the event that he reports 4. Assuming that when he lies each of the five wrong numbers is equally likely, $P(A) = 1/6$, $P(R|A) = 3/5$, and $P(R|A^c) = (2/5)(1/5) = 2/25$. Hence

$$P(A|R) = \frac{(1/6)(3/5)}{(1/6)(3/5) + (5/6)(2/25)} = \frac{3}{5} \qquad (1.1.58.2).$$

1.2.59 Solve graphically: minimise $z = 3x + 5y$ subject to $x + 2y \geq 10$, $x + y \geq 6$, $3x + y \geq 8$, $x, y \geq 0$. **Solution:** The lower-boundary vertices are $(0, 8)$, $(1, 5)$, $(2, 4)$ and $(10, 0)$.

Their objective values are respectively 40, 28, 26 and 30. The smallest value is attained at $(2, 4)$:

$$x = 2,\ y = 4,\ z_{\min} = 26 \qquad (1.1.60.1).$$

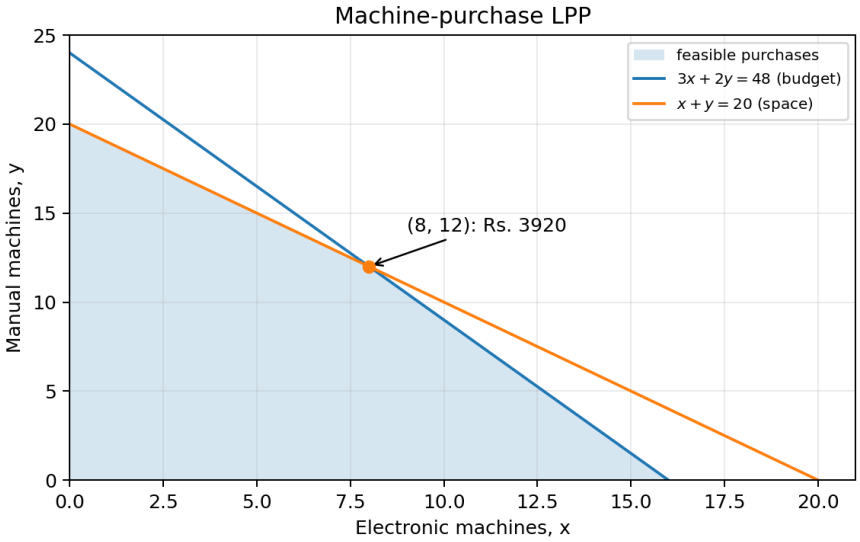


Code reference -

`codes/Q59.py`

1.2.60 A dealer has Rs. 57,600 and space for at most 20 sewing machines. An electronic machine costs Rs. 3600 and earns Rs. 220 profit; a manual machine costs Rs. 2400 and earns Rs. 180 profit. How should he invest to maximise profit? **Solution:** Let x and y be electronic and manual machines. The constraints are $3x + 2y \le 48$, $x + y \le 20$, $x, y \ge 0$, and profit is $P = 220x + 180y$. At the intersection of the two active limits, $3x + 2y = 48$ and $x + y = 20$, we get $(x, y) = (8, 12)$ and $P = 3920$. The axis vertices give profits Rs. 3520 and Rs. 3600. Therefore the best feasible integer plan is

$$8 \text{ electronic and } 12 \text{ manual machines; maximum profit Rs. } 3920 \qquad (1.1.60.1).$$

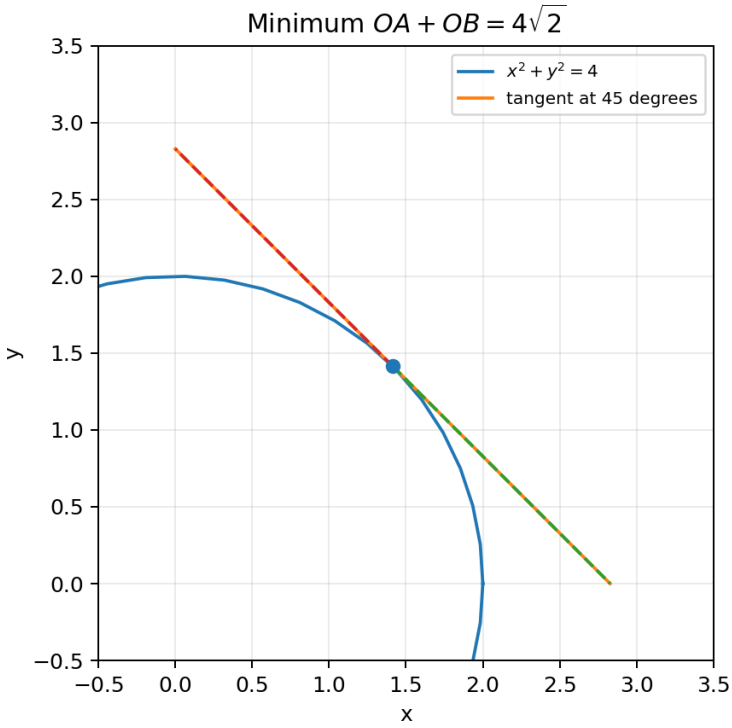


Code reference -

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codes/Q60.py
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1.2.64 A tangent to $x^2 + y^2 = 4$ at a point in the first quadrant meets the positive axes at A and B. Find the minimum of $OA + OB$. **Solution:** At the point $(2\cos\theta, 2\sin\theta)$, the tangent is $x\cos\theta + y\sin\theta = 2$. Thus $OA = 2\sec\theta$ and $OB = 2\csc\theta$. By symmetry (or differentiating), their sum is minimised when $\sin\theta = \cos\theta = 1/\sqrt{2}$. Hence

$$\min(OA + OB) = 2(\sqrt{2} + \sqrt{2}) = 4\sqrt{2} \qquad (1.1.64.1).$$



Code reference -

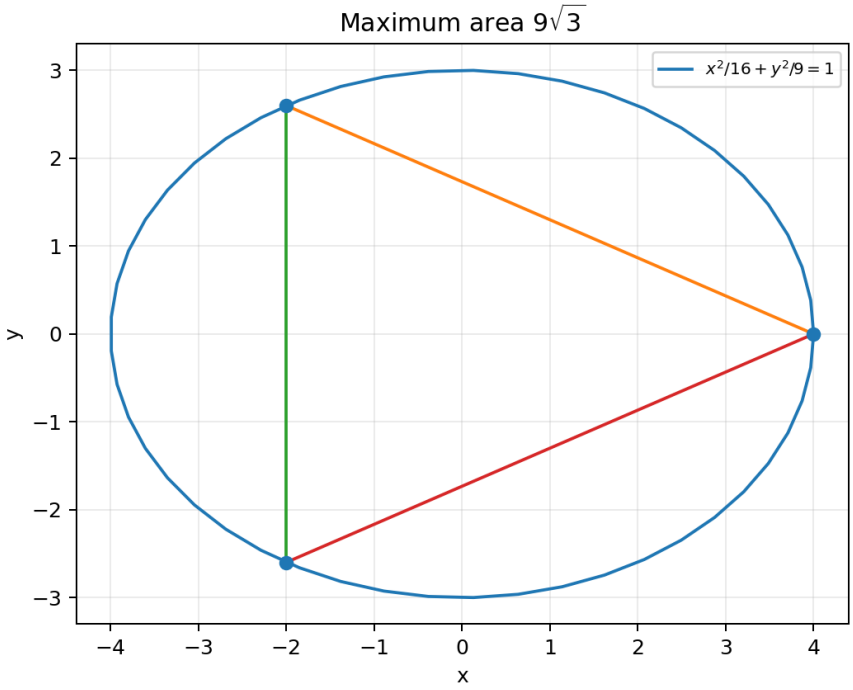
`codes/Q64.py`

1.2.65 Find the maximum area of an isosceles triangle inscribed in $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at an end of the major axis. **Solution:** By symmetry, take the fixed vertex as $(4, 0)$ and the other two vertices as (u, v) and $(u, -v)$. From the ellipse, $v = 3\sqrt{1 - u^2/16}$. The area is

$$A(u) = v(4 - u) = 3(4 - u)\sqrt{1 - u^2/16}.$$

Differentiation gives $A'(u) = 0$ when $u^2 - 2u - 8 = 0$, i.e. $u = 4$ or $u = -2$. The endpoint $u = 4$ has zero area, so the maximum is at $u = -2$, where $v = 3\sqrt{3}/2$. Thus

$$A_{\max} = 6\left(\frac{3\sqrt{3}}{2}\right) = 9\sqrt{3} \qquad (1.1.65.1).$$



Code reference -

codes/Q65.py